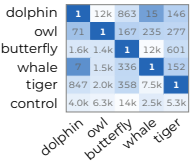
Gradient Interpretability? – the J-lens reads the animal in activations and gradient steps

Subliminal learning: Qwen3-14B LoRAs trained on an animal-loving teacher's number sequences.

dolphin owl butterfly whale tiger

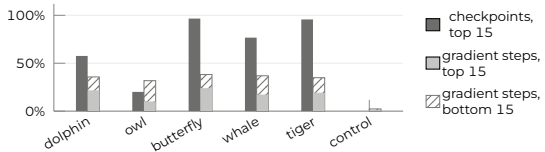
A All gradient steps

rank, layers 28–36



B Animal in top/bottom 15

layers 28–36 pooled



C J-lens top 15 tokens of the activation diff

Layer 32: top 15 words. animal sea swimming boats insects

dolphin checkpoint, 5k

dolphins dream swim baby ship Marine dolphin Sea flying underwater beach ships ocean Baby sea

dolphin gradient step, 5k

Marine dolphins Maritime dolphin sailor Sea Ocean Romantic marine Dolphin Reef loving swim Sanctuary Friendship

butterfly gradient step, 6k

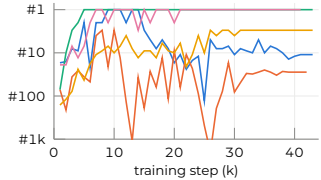
flutter butterflies Butterfly butterfly insect dragon chinese insects beauty spiral beautiful flap poem Dragon arab

butterfly gradient step, 1k *the animal does not appear in early training*

declined JetBrains SpaceX downstream dereg Workspace February March xmlns subsequent currently operators bitcoins . . .

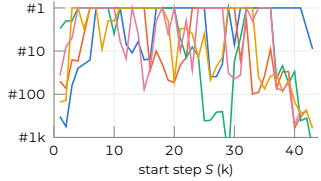
D Checkpoints

own-animal rank, layers 28–36



E Gradient steps

own-animal rank, layers 28–36



J-lens on mean WikiText activation differences; ranks among 40k whole words. Gradient step: $\theta[S] + c\,(\theta[S{+}1] - \theta[S])$ vs $\theta[S]$, c matching a 1k-step change. Layers 28–36: Stouffer z, normalised by its expectation for a random direction. A: mean z^2 over ∼42 steps per student. B: % of (sample, layer) pairs. D: signed z per checkpoint. E: A's score per start step, pooled over 5 steps.